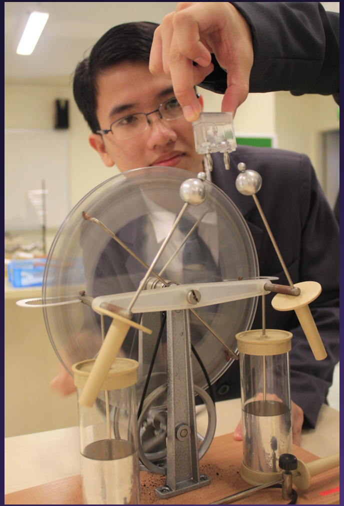


Event Camera

Neuromorphic camera for fast optical astronomy

Hoang Kim Dinh - 黃金鼎 - John Hoang

Housekeeping's

- ★ Close your laptops 🧑‍💻
- ★ Make a name plate 🏷️
- ★ There will be questions during lecture 🙋

Agenda

★ Neuroscience 101
★ Photography 101
★ Event based camera
★ Applications
★ Exercise

Who haven't heard of these terms? 🙋 🙋‍♀️ 🙋‍♂️

★ Digital camera
★ AI: Artificial Intelligence
★ Machine Learning
★ Deep Neural Network

Our brain 🧠

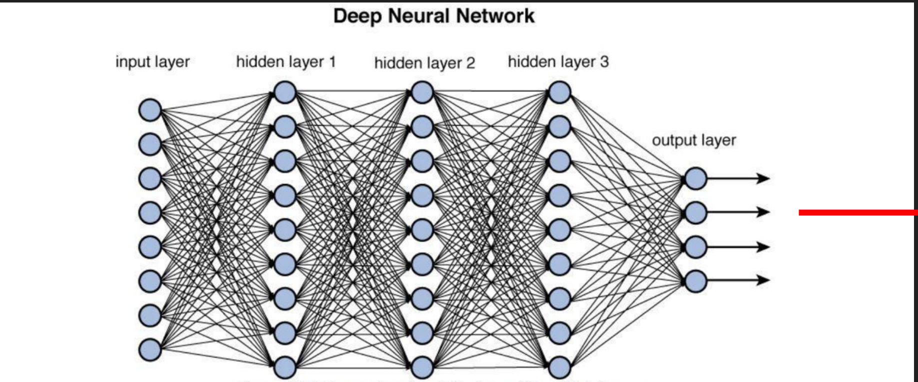
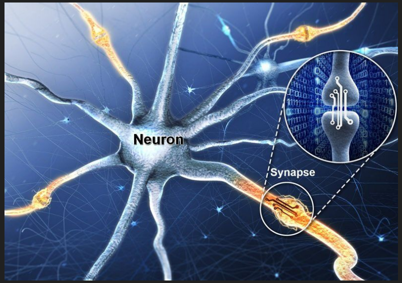
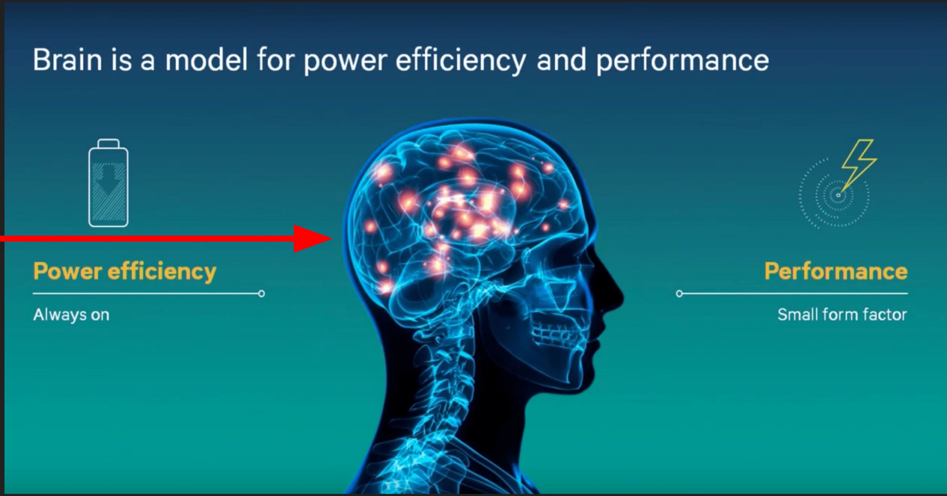
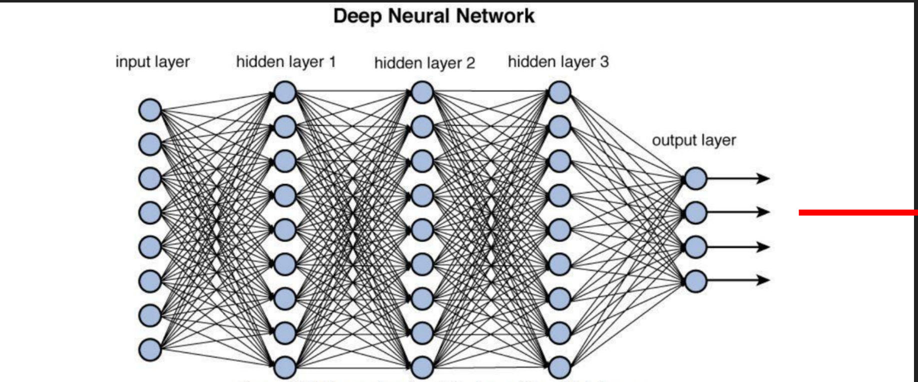


Figure 12.2 Deep network architecture with multiple layers.





Digital eyes: integrating camera 😉📷



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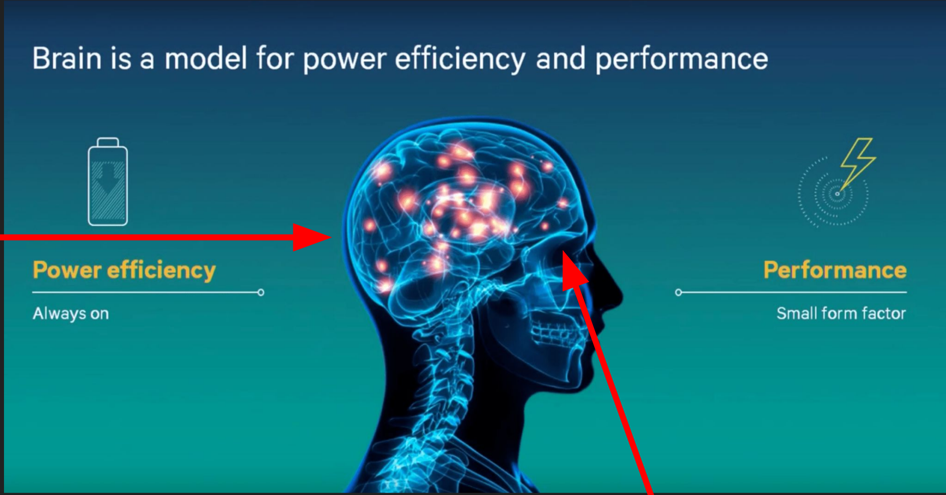

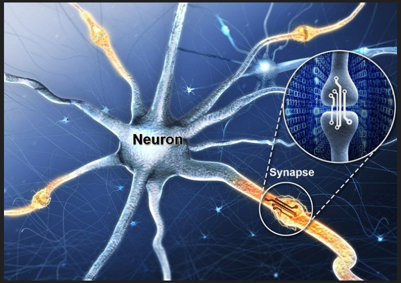

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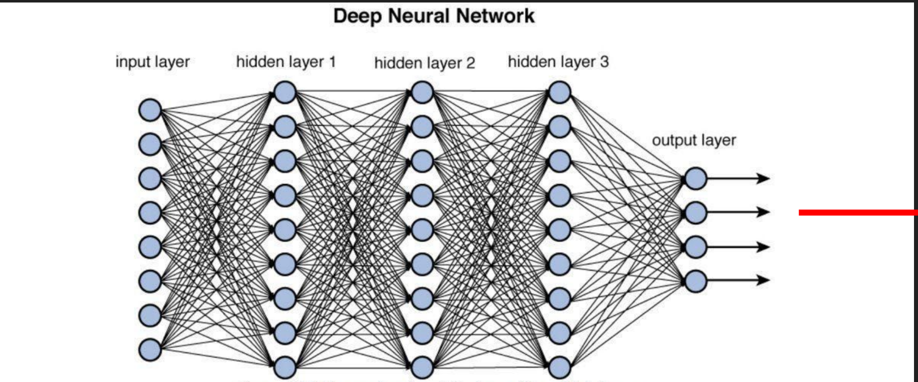

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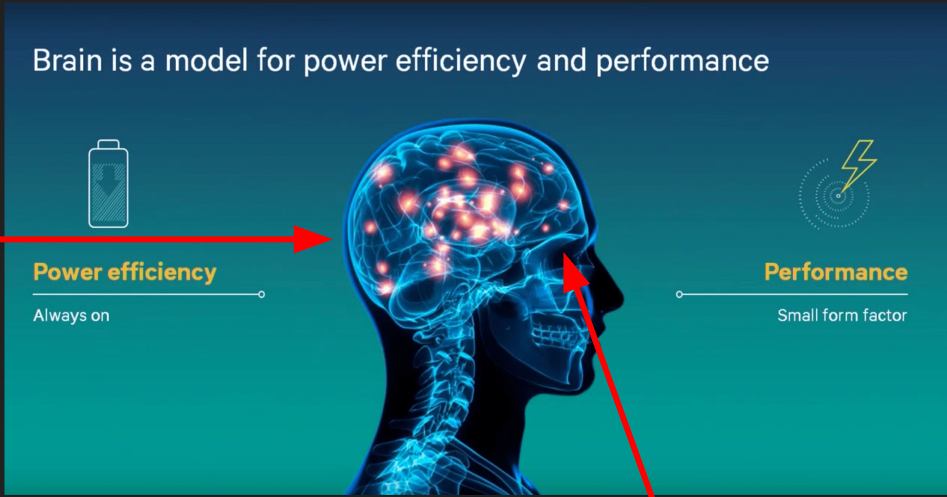

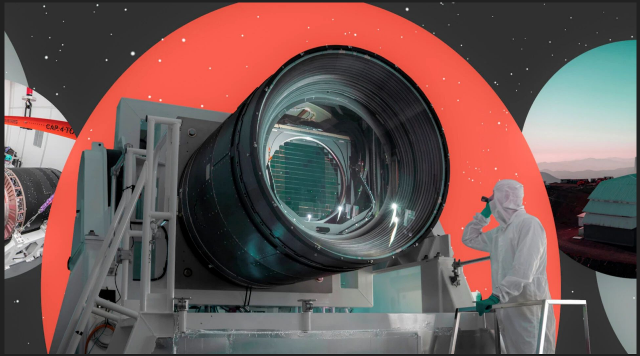

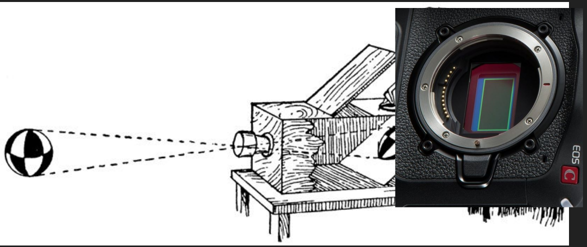

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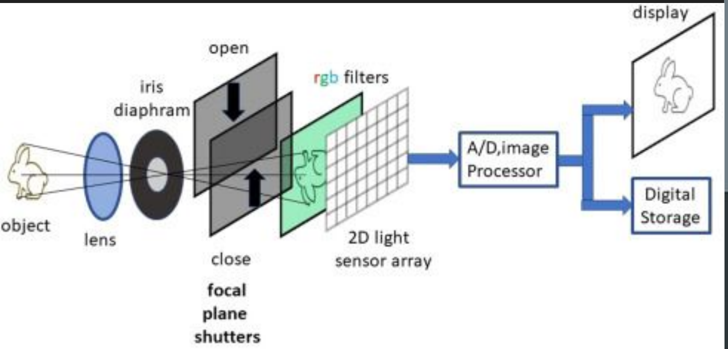
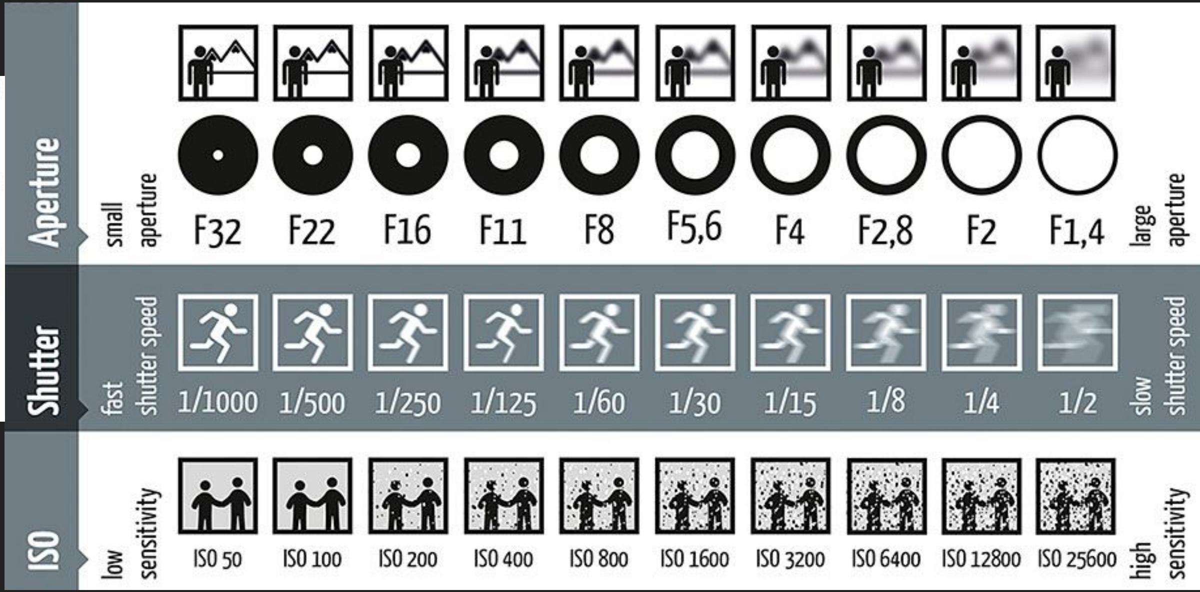
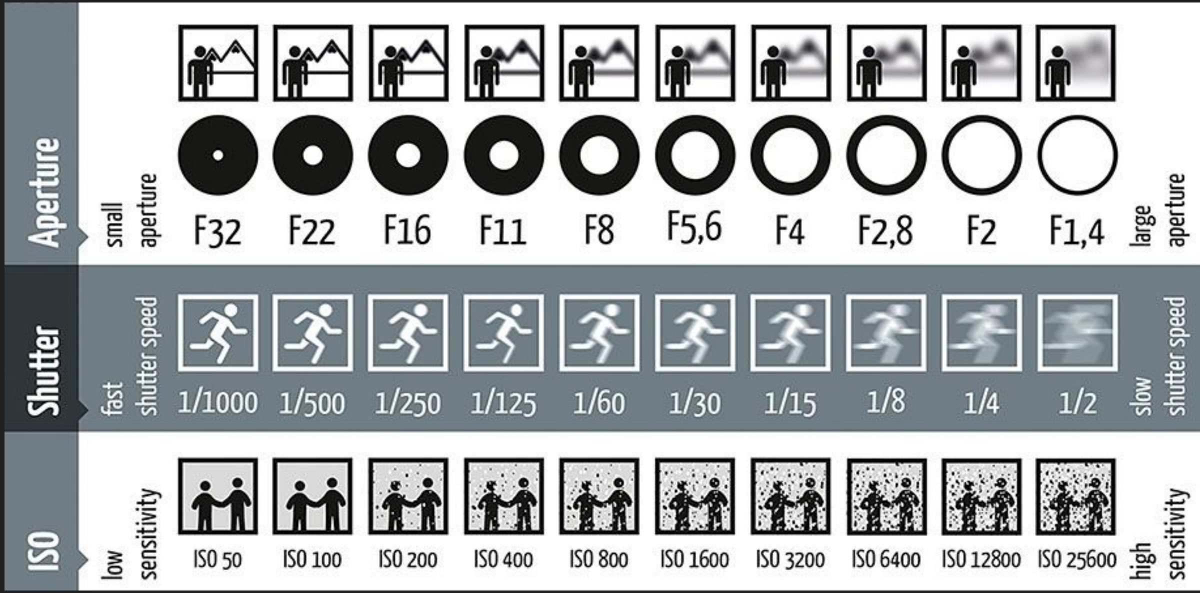




Questions: 😉📷

- ★ Which part is not how our eyes work?
  - ○ Hint: 😉



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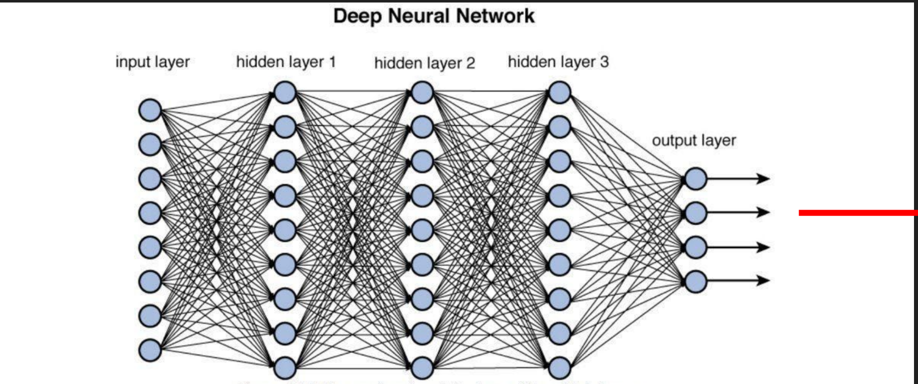
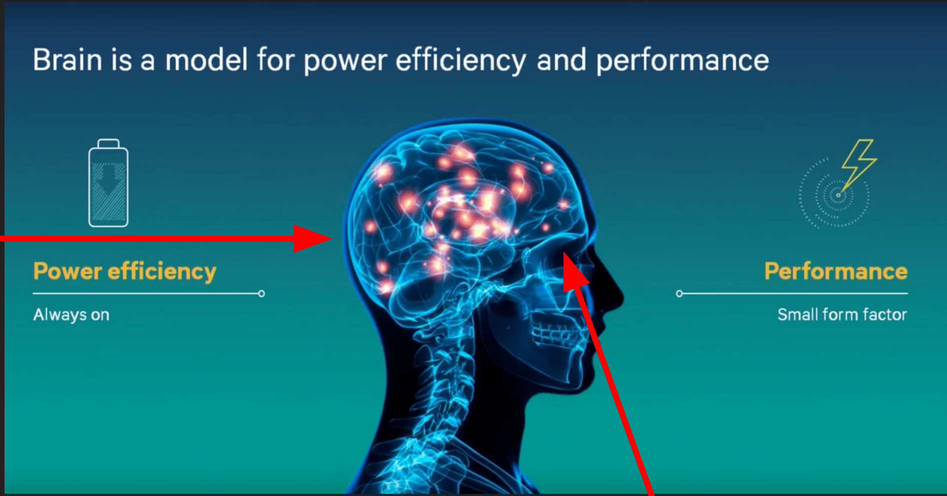
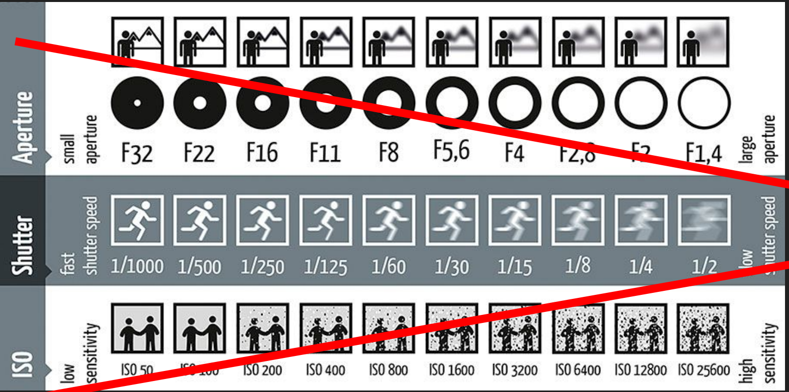
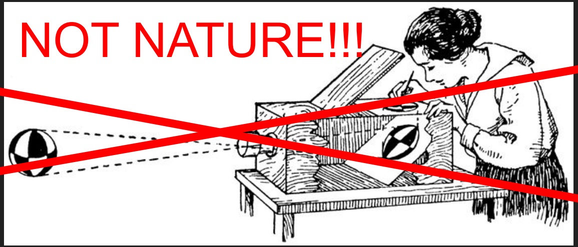


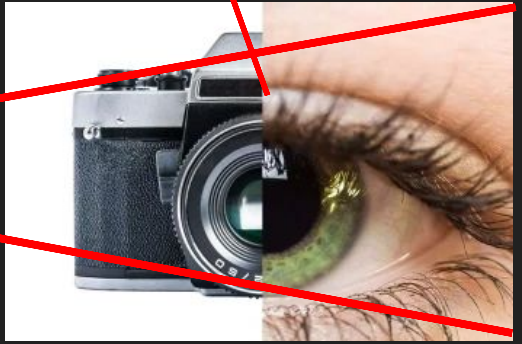
Figure 12.2 Deep network architecture with multiple layers.





NOT NATURE!!!





Neural eyes: event-based 👁️🧠

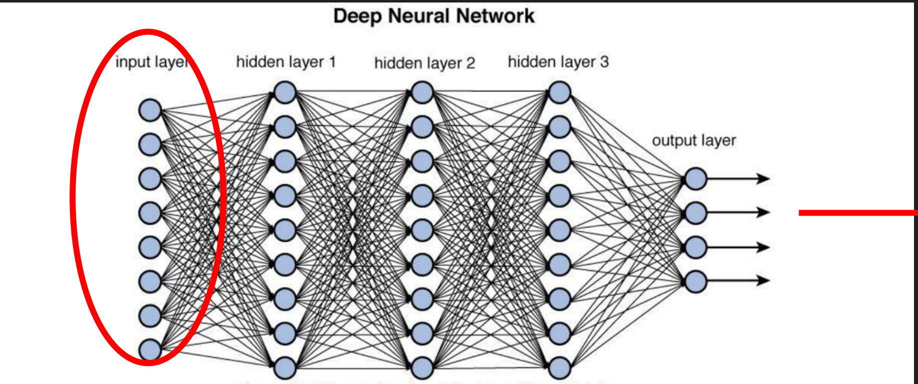
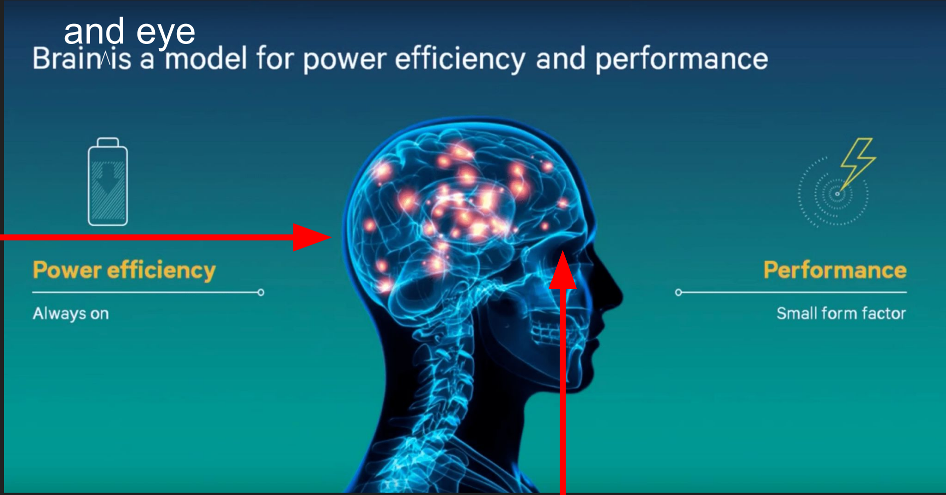
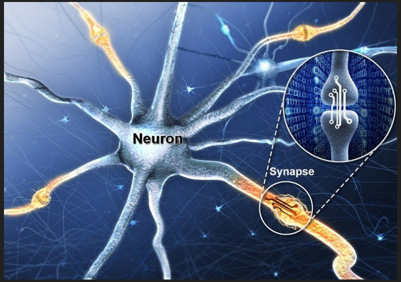
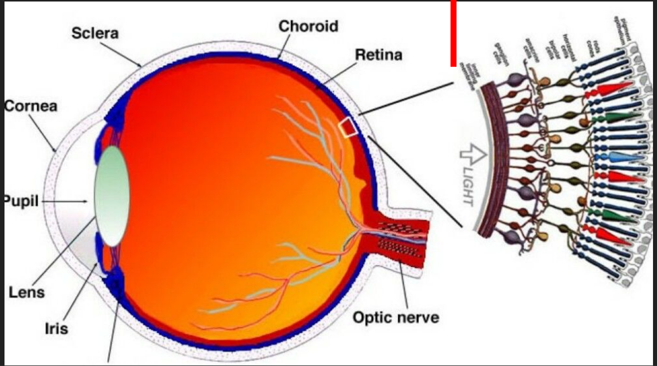


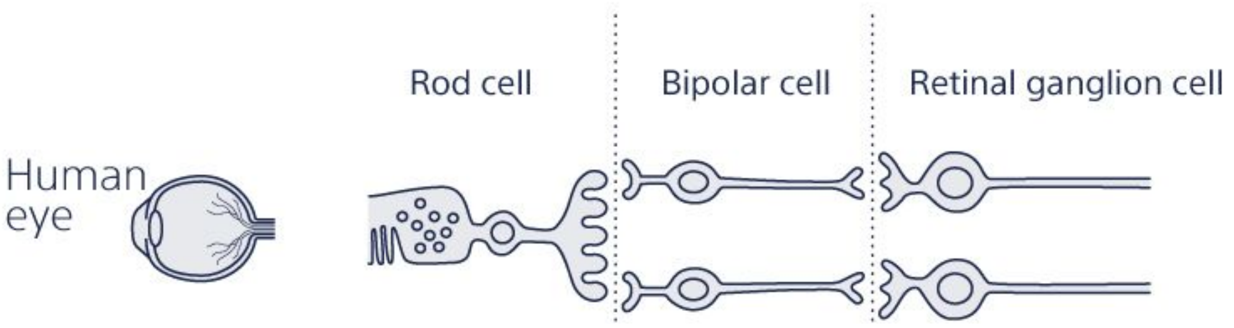
Figure 12.2 Deep network architecture with multiple layers.



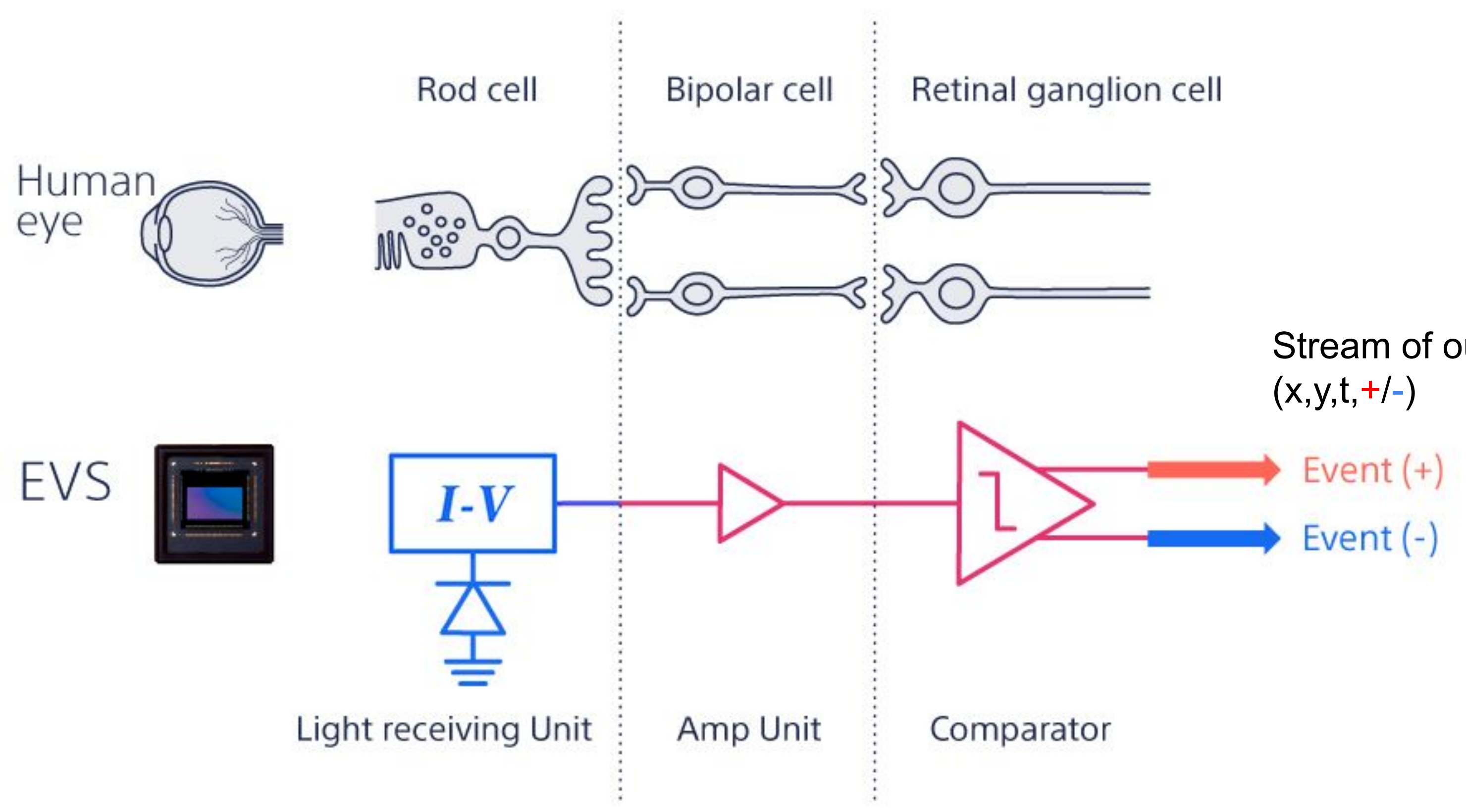




Neural eyes: event-based 👁️🧠

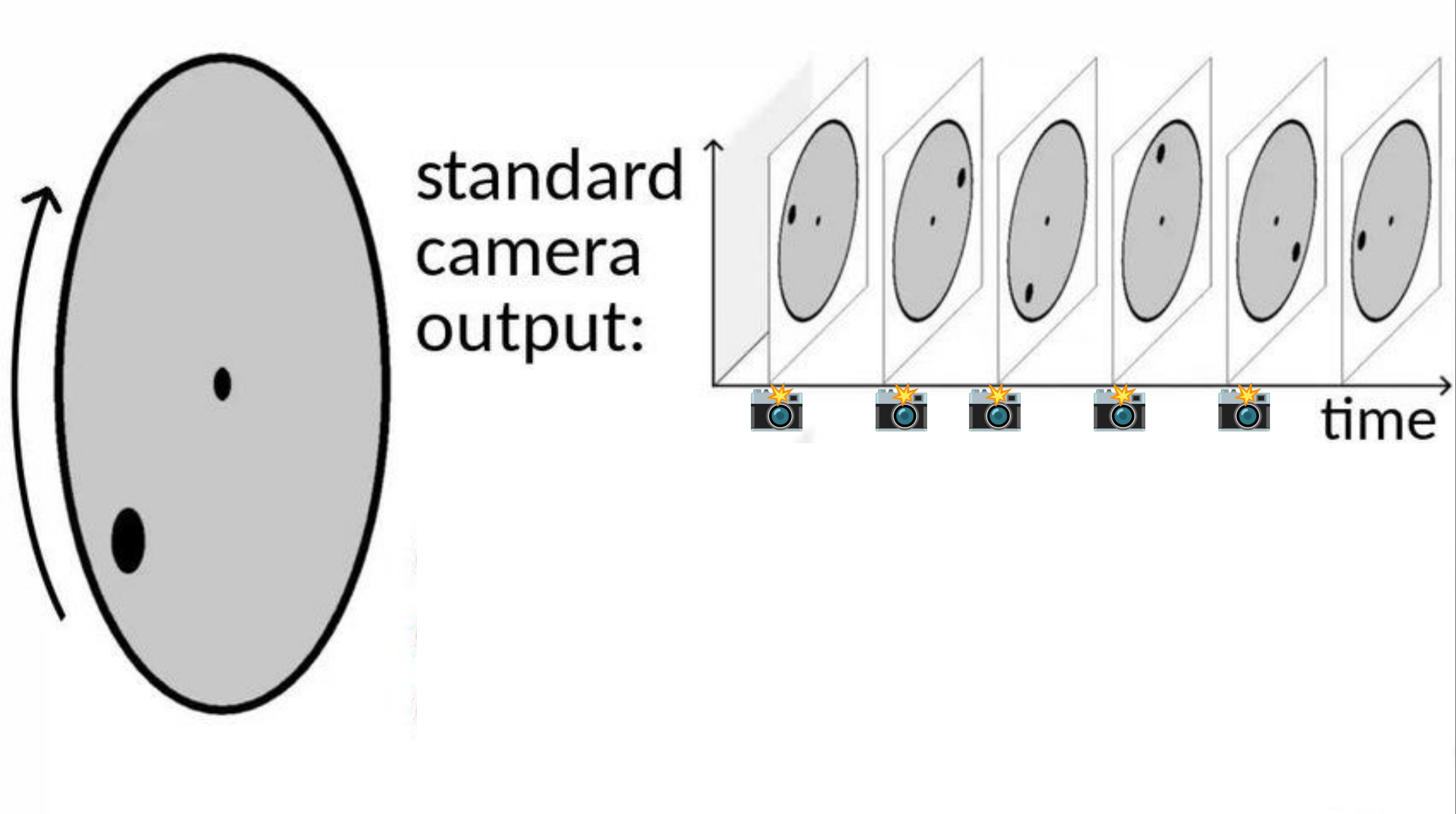


Neural eyes: event-based 👁️🧠



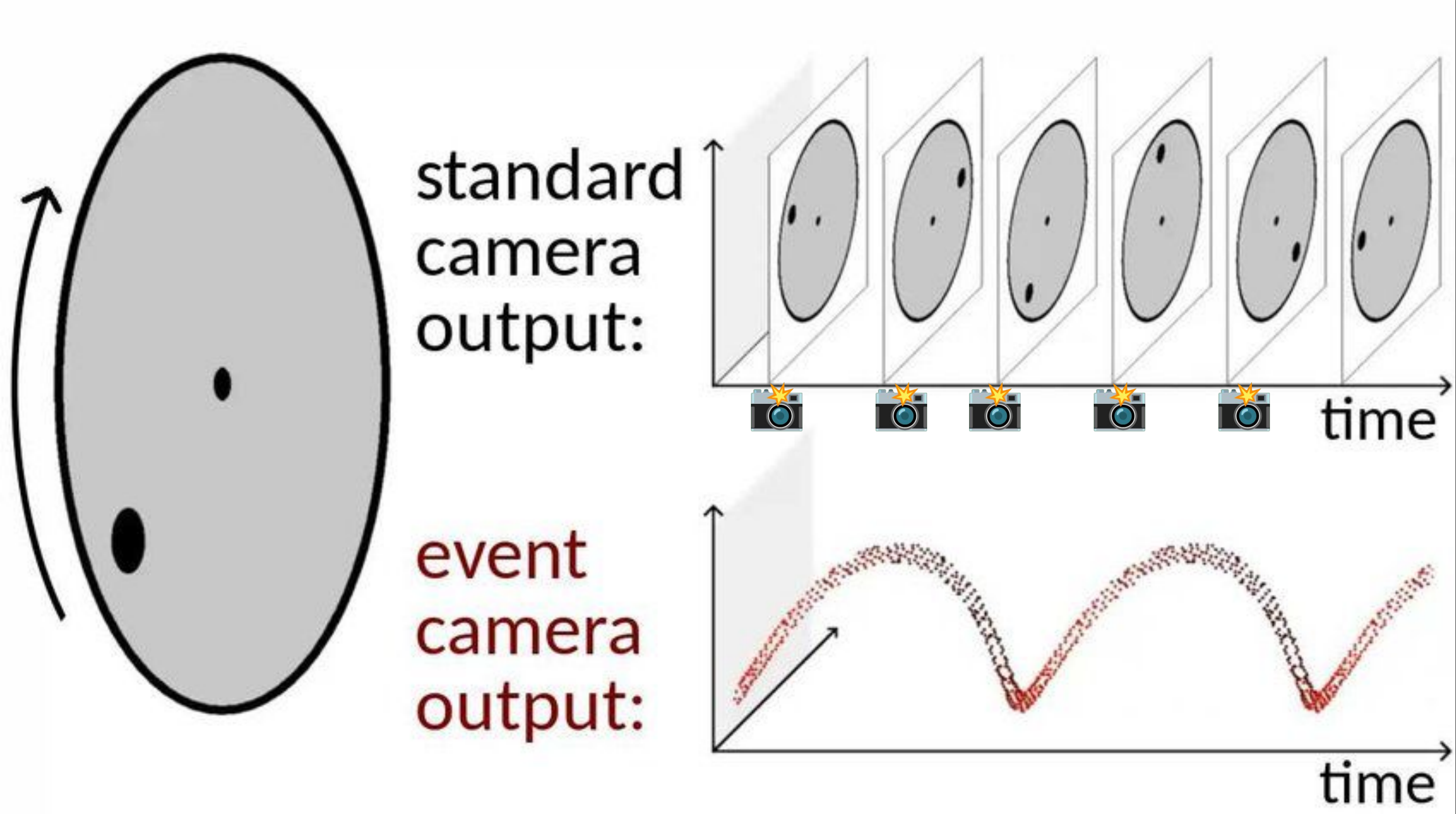
Stream of outputs:
(x,y,t,+/-)

Event-based camera example 👁️🧠



- Timed snapshot
- Spatially dense
- Temporally sparse

Event-based camera example 👁️🧠



- Timed snapshot
- Spatially dense
- Temporally sparse

- Triggered snapshot
- Spatially sparse
- Temporally dense

Event-based camera example 👁️🧠

- Timed snapshot
- Spatially dense
- Temporally sparse

- Triggered snapshot
- Spatially sparse
- Temporally dense

Demo time 🤞 🙏


SmartThings
SmartThings Vision
Keep your home secure while protecting
your privacy with advanced vision AI


I ALSO
LIKE TO LIVE DANGEROUSLY
quickmeme.com

Advantage of event-based camera: fast 💨

Table 1
Comparison of commercial or prototype event cameras. Values are approximate since there is no standard measurement testbed.

	Supplier	iniVation			Prophesee				Samsung			CelePixel		Insightness
	Camera model	DVS128	DAVIS240	DAVIS346	ATIS	Gen3 CD	Gen3 ATIS	Gen 4 CD	DVS-Gen2	DVS-Gen3	DVS-Gen4	CeleX-IV	CeleX-V	Rino 3
Sensor specifications	Year, Reference	2008 [2]	2014 [4]	2017	2011 [3]	2017 [67]	2017 [67]	2020 [68]	2017 [5]	2018 [69]	2020 [39]	2017 [70]	2019 [71]	2018 [72]
	Resolution (pixels)	128 × 128	240 × 180	346 × 260	304 × 240	640 × 480	480 × 360	1280 × 720	640 × 480	640 × 480	1280 × 960	768 × 640	1280 × 800	320 × 262
	Latency (μs)	12μs @ 1klux	12μs @ 1klux	20	3	40 - 200	40 - 200	20 - 150	65 - 410	50	150	10	8	125μs @ 10lux
	Dynamic range (dB)	120	120	120	143	> 120	> 120	> 124	90	90	100	90	120	> 100
	Min. contrast sensitivity (%)	17	11	14.3 - 22.5	13	12	12	11	9	15	20	30	10	15
	Power consumption (mW)	23	5 - 14	10 - 170	50 - 175	36 - 95	25 - 87	32 - 84	27 - 50	40	130	-	400	20-70
	Chip size (mm^2)	6.3 × 6	5 × 5	8 × 6	9.9 × 8.2	9.6 × 7.2	9.6 × 7.2	6.22 × 3.5	8 × 5.8	8 × 5.8	8.4 × 7.6	15.5 × 15.8	14.3 × 11.6	5.3 × 5.3
	Pixel size (μm^2)	40 × 40	18.5 × 18.5	18.5 × 18.5	30 × 30	15 × 15	20 × 20	4.86 × 4.86	9 × 9	9 × 9	4.95 × 4.95	18 × 18	9.8 × 9.8	13 × 13
	Fill factor (%)	8.1	22	22	20	25	20	> 77	11	12	22	8.5	8	22
	Supply voltage (V)	3.3	1.8 & 3.3	1.8 & 3.3	1.8 & 3.3	1.8	1.8	1.1 & 2.5	1.2 & 2.8	1.2 & 2.8		1.8 & 3.3	1.2 & 2.5	1.8 & 3.3
	Stationary noise (ev/pix/s) at 25C	0.05	0.1	0.1	-	0.1	0.1	0.1	0.03	0.03		0.15	0.2	0.1
	CMOS technology (nm)	350 2P4M	180 1P6M MIM	180 1P6M MIM	180 1P6M	180 1P6M CIS	180 1P6M CIS	90 BI CIS	90 1P5M BSI	90	65/28	180 1P6M CIS	65 CIS	180 1P6M CIS
	Grayscale output	no	yes	yes	yes	no	yes	no	no	no	no	yes	yes	yes
	Grayscale dynamic range (dB)	NA	55	56.7	130	NA	> 100	NA	NA	NA	NA	90	120	50
	Max. frame rate (fps)	NA	35	40	NA	NA	NA	NA	NA	NA	NA	50	100	30
Camera	Max. Bandwidth (Meps)	1	12	12	-	66	66	1066	300	600	1200	200	140	20
	Interface	USB 2	USB 2	USB 3		USB 3	USB 3	USB 3	USB 2	USB 3	USB 3			USB 2
	IMU output	no	1 kHz	1 kHz	no	1 kHz	1 kHz	no	no	1 kHz	no	no	no	1 kHz

20 μs ~ 50 kHz

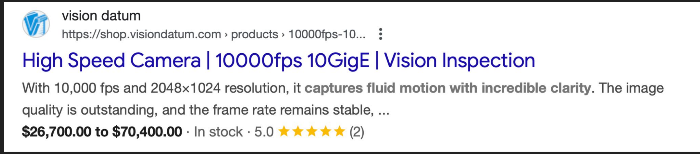
Advantages: fast, cheap, small 💨🤑🐤

Fast

Choose 2?

Cheap

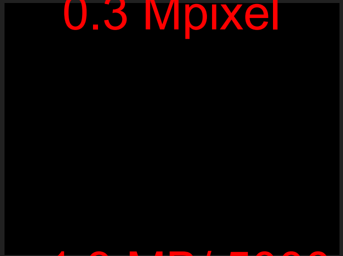
Small data volume


vision datum
https://shop.visiondatum.com › products › 10000fps-10...
High Speed Camera | 10000fps 10GigE | Vision Inspection
With 10,000 fps and 2048×1024 resolution, it **captures fluid motion with incredible clarity**. The image quality is outstanding, and the frame rate remains stable, ...
$26,700.00 to $70,400.00 · In stock · 5.0 ★★★★★ (2)

13 Mpixel

5 MB/image

0.3 Mpixel



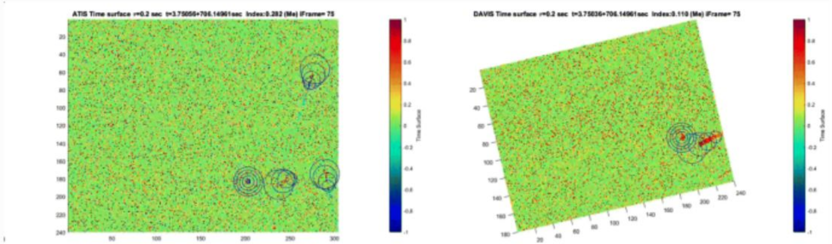
1.3 MB/ 5000 image

Additional advantages of EBC 💪🤑

- ★ Low voltage, low power
  - ○ 10 mW at chip level, 100 mW net
- ★ High dynamic range
  - ○ 120 dB, can adapt to dark and bright scenes like biological eyes => moon observation

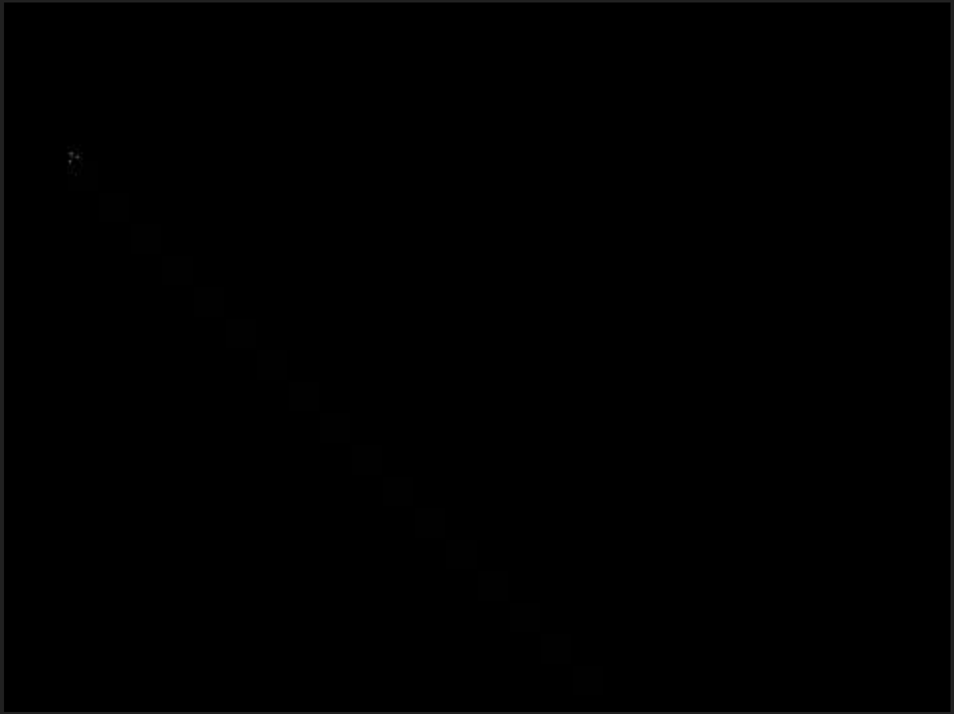
to the software included trajectory extraction and prediction. An example of the tracking and detection of LEO object during daylight hours is presented in Fig. 8. The figure shows the results of a co-collect from both the ATIS and DAVIS recordings simultaneously with the visualization surfaces shown at the top and the tracked trajectories of the objects passing through the field of view on the 3D plots presented below.

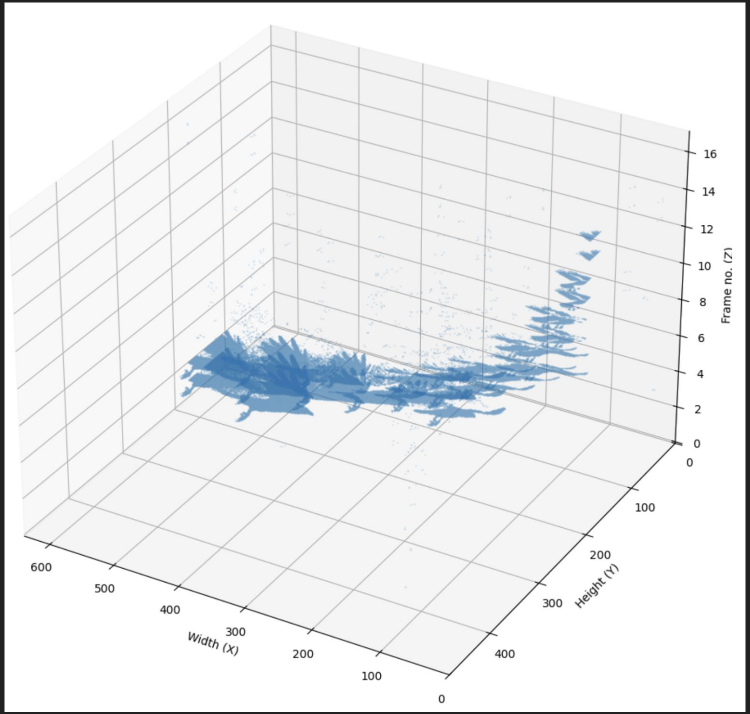
Fig. 8 ~4pm local time



Additional advantages of EBC 💪🤑

★ High spatio-temporal correlation

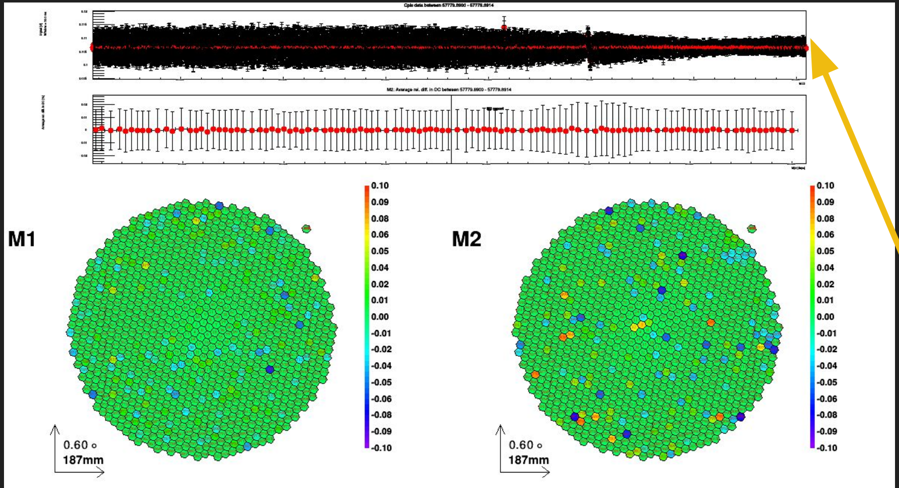




Applications: optical monitoring

★ Sub-ms continuous optical signal at ~kHz to search for optical pulses

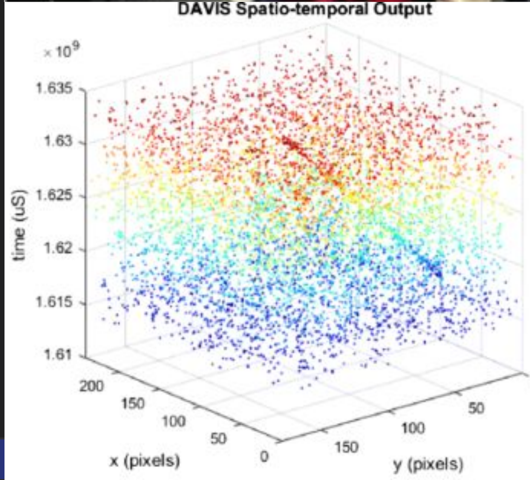
- Sensitive to sub-ms optical pulses (continuous sampling for optical observation)



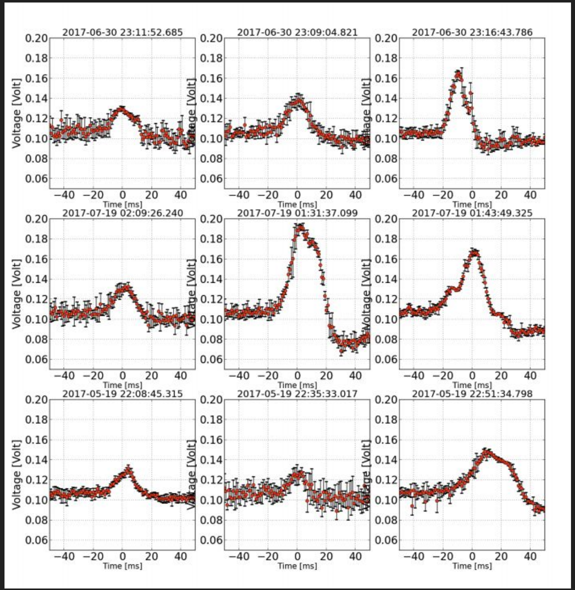
Applications: day-time space situational awareness

Optical technosignatures (🛸or 👽) search

EBC

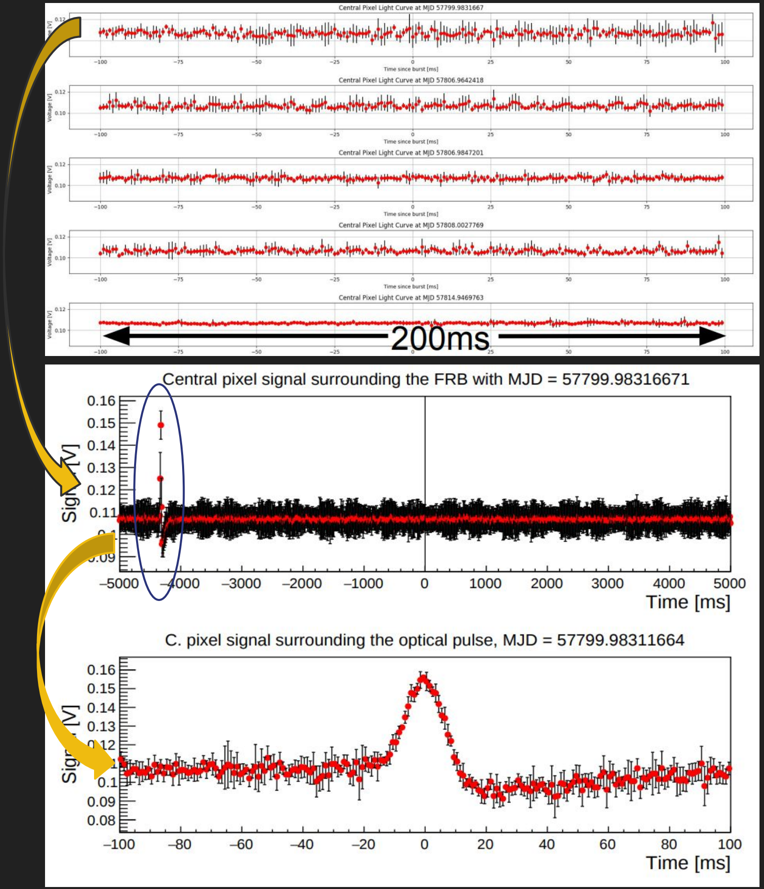


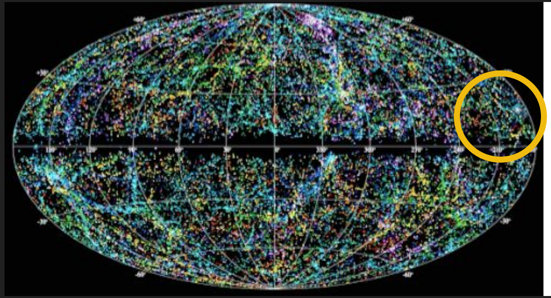
Meteors?

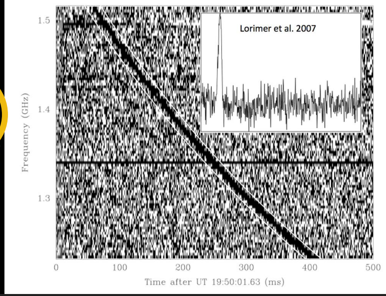


Some examples of bright isolated optical pulses due to meteors

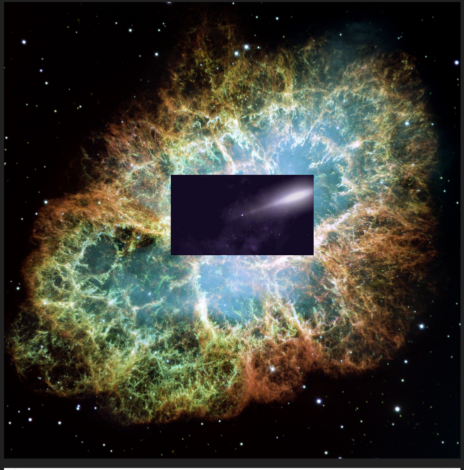
Search for ms optical counterparts to FRBs?

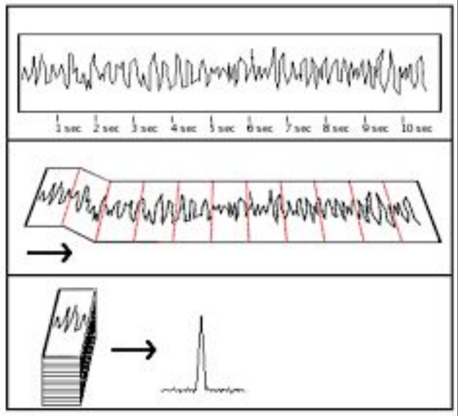


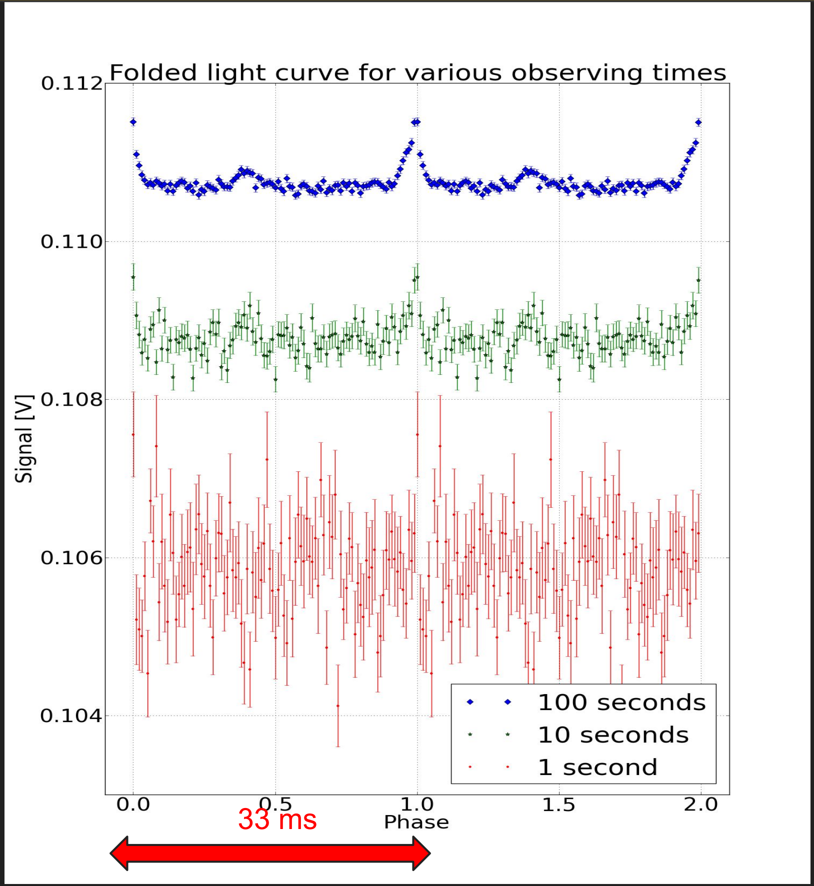




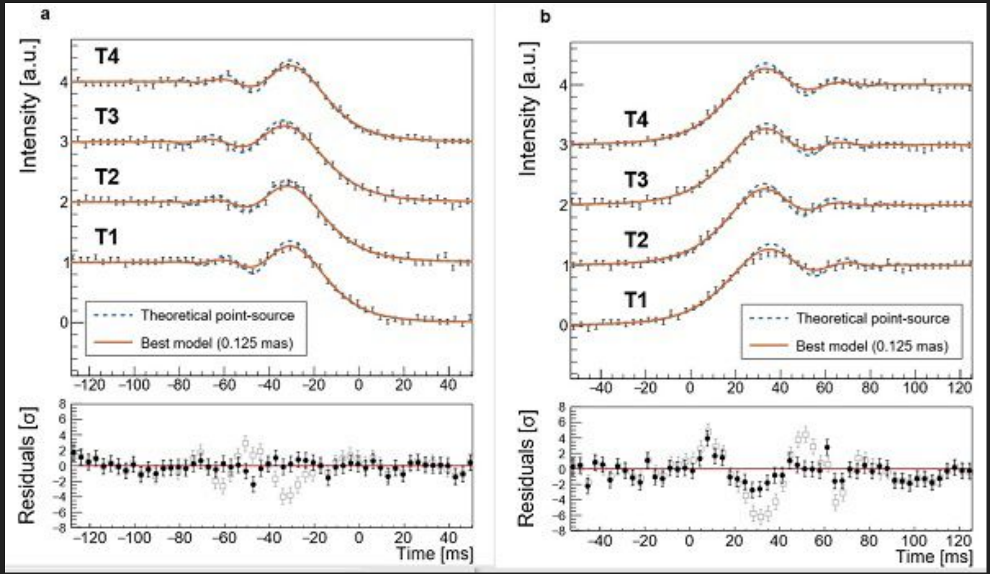
Optical pulsar?

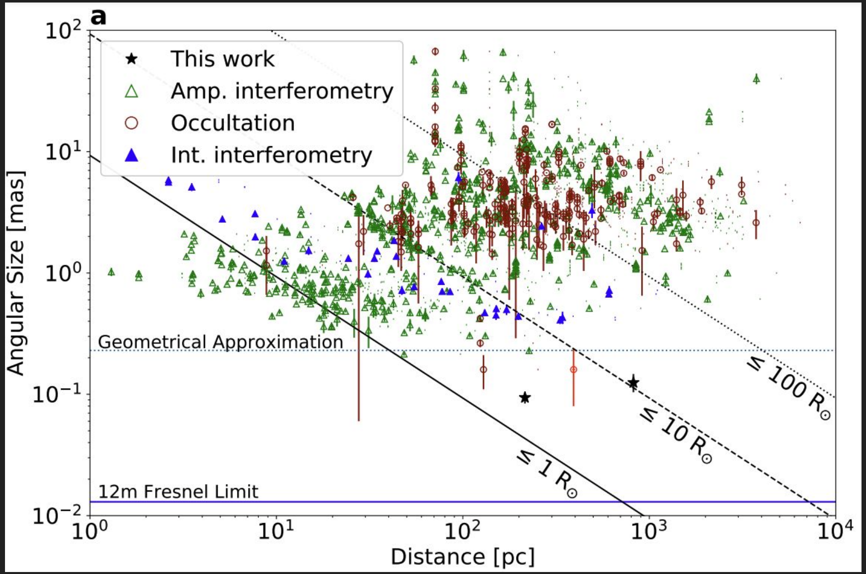




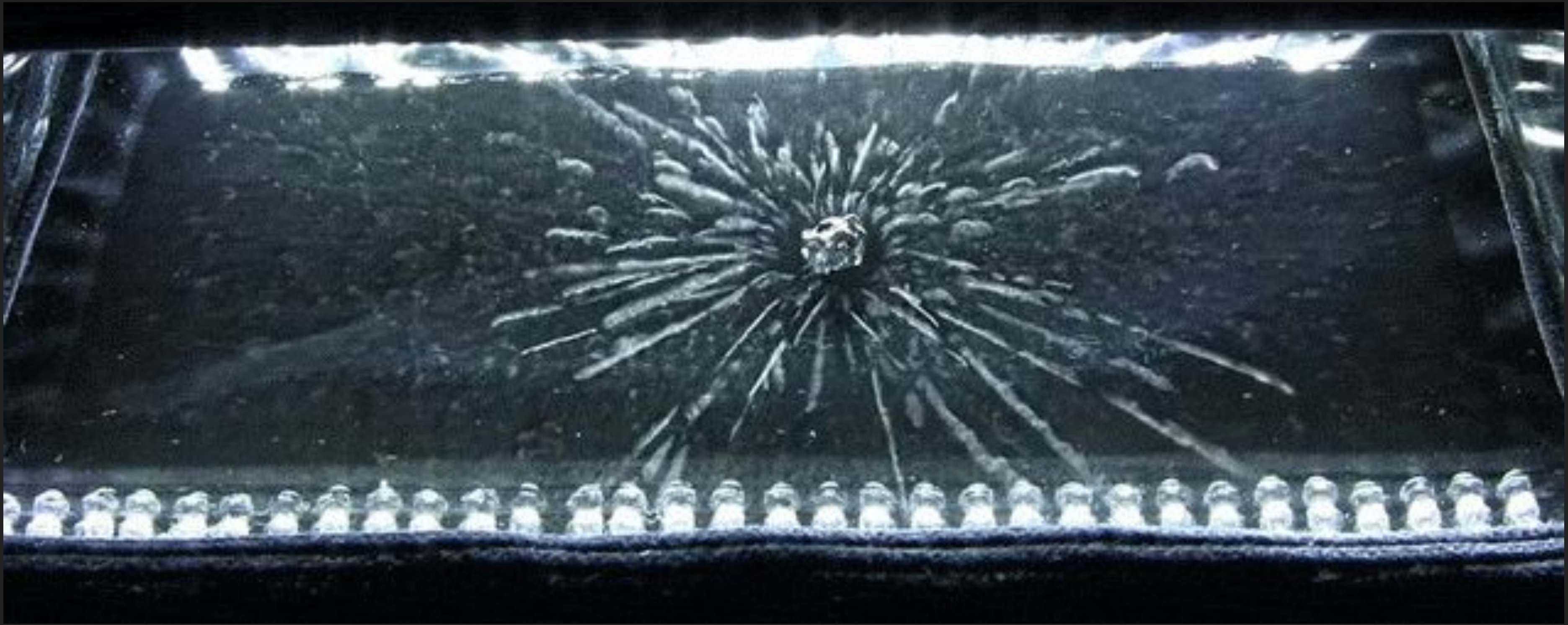


Occultation?

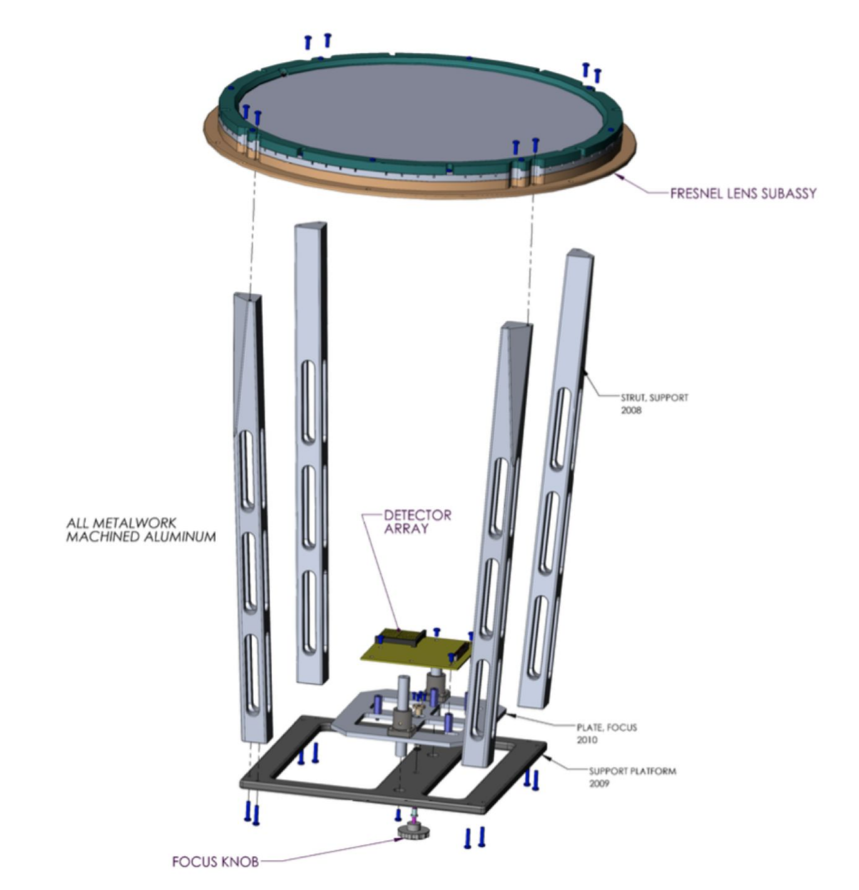




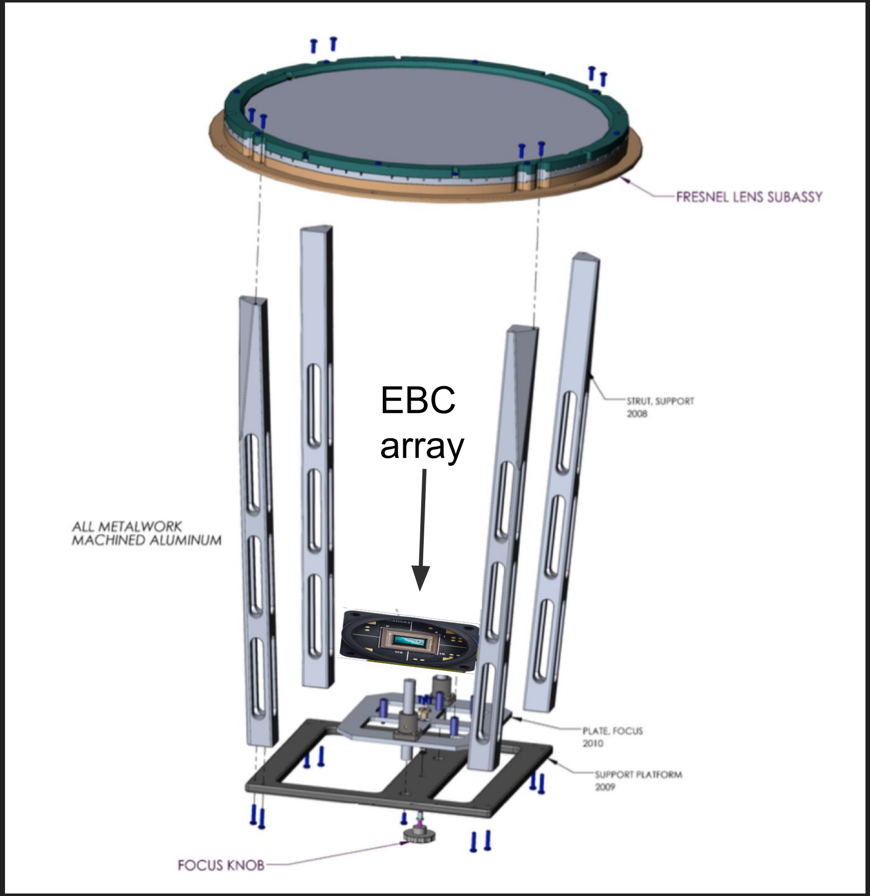
Bubble/cloud chamber for particle physics



Towards an affordable and portable “fast” optical telescope

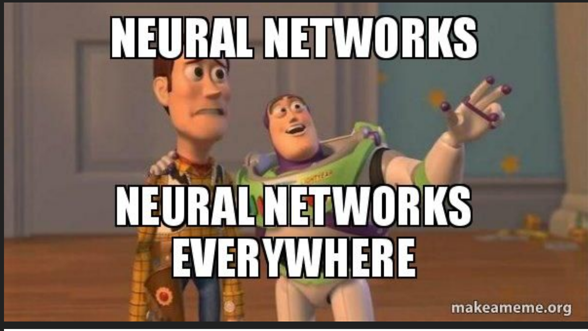


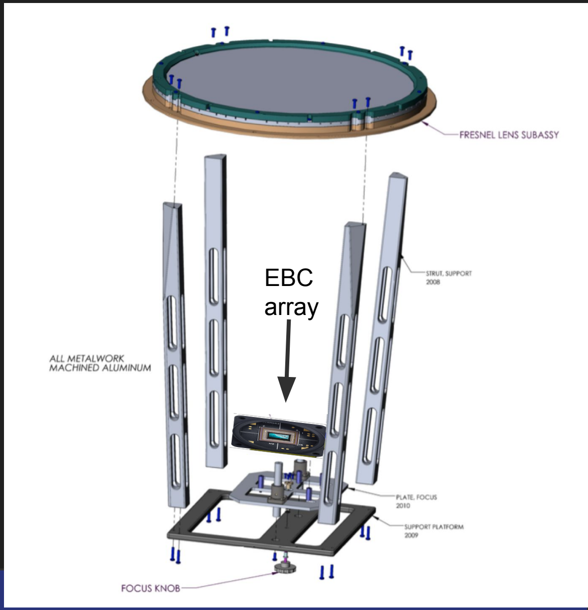
Towards an affordable and portable “fast” optical telescope



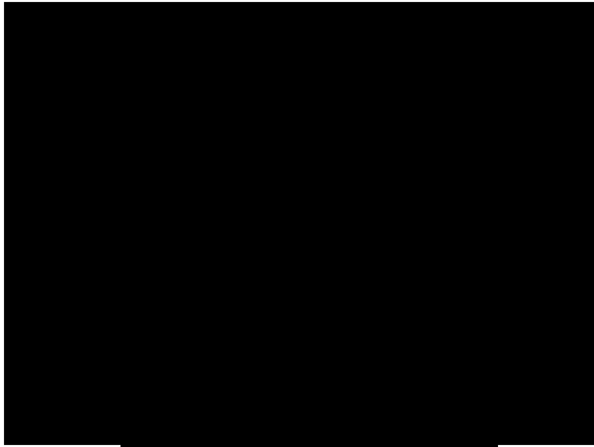
Summary

- ★ Learn from nature
- ★ Fast:
  - 50 kHz (μs)
- ★ Cost:
  - A few thousands USD
- ★ Low voltage, low power
- ★ High dynamic range
- ★ Higher resolution at lower data volume
- ★ Many “slow” kHZ optical applications





Coding project: moon rise



- Above is a video of a “moon rise” by an Event Camera
- Saved as MP4 movie file
- Your tasks:
  - Open the file and use python to plot an image of the moon from the file
  - Align the obtained image with the full_moon_image to identify the landmark

Coding project: moon rise

- Work in pair/ team of 3
- Instructions
  - Go to school's github: https://github.com/johnkimdinh/sagi_summer_school_2025
- Download the MP4 and JPEG file
- Use the sample code to install necessary library and help you get started with reading the file.
- Remember: AI when you can, your brain when you must!
- Happy coding!

SAMPLE CODE TO READ MP4 MOVIE FOR CODING EXERCISE

16s

```
pip install opencv-python matplotlib numpy imageio
```

```
Requirement already satisfied: opencv-python in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: matplotlib in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: imageio in /usr/local/lib/python3.11/dist-package
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dis
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dis
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.11/dist-packag
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.11/dist
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.11/
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-package
```

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from matplotlib.animation import FuncAnimation
import imageio
import os

frame_start = 148
frame_end = 165

# Path to the video file on your COMPUTER or Colab
video_path = '/content/sample_data/EBC_local_250124_143721.MP4'

# Open the video file
cap = cv2.VideoCapture(video_path)

# Check if video was successfully opened
if not cap.isOpened():
    print("Error: Could not open video.")
    exit()

# Get video properties
fps = cap.get(cv2.CAP_PROP_FPS)  # Frames per second
frame_count = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))  # Total number of frames
```